

BRIEF:

The lap time of a race car is governed by many factors including the acceleration rate, braking rate, road geometry, road friction (tendency to skid) and centre of gravity (tendency to roll over). Given a specific path on the road, it is possible to combine these factors to create a physical model that predicts the maximum speed at which you can drive at each point on the road. There are several different approaches that can be taken to solve the problem of finding the optimum path.

The following approach based on a Genetic Algorithm might be considered.

A suitable path can be created by fitting splines (or Bezier curves) to the centre line path describing a track. Using the physical model of the car and track it is possible to create and estimate of lap times. The track has width, so the car has many options of the path to take around the track. In this case, the track can be divided into several connected segments each with an entry point and exit point, represent as a horizontal position on the road. Using a genetic algorithm the optimum path around the track should be identifiable.

The steps for this project could include:

- 1: Choosing a development platform, (e.g. Processing, PyGame, OpenGL, WebGL).
- 2: Defining the track as a set of way points (generated synthetically or extracted from a digital map such as OpenStreetMap).
- 3: Interpolate a set of points between the points using a cubic spline fit.
- 4: Calculate the maximum velocity a car can drive at each point on the road (by equating the centripetal force with the road friction force).
- 5: Model the acceleration and deceleration of the car on the road using the equations of straight-line motion.
- 6: Calculate the lap time.
- 7: Extend the model from single track to a road with width.
- 8: Generate multiple paths around the track
- 9: Use a genetic algorithm to find the optimal path around the track.

This project might suit a competent programmer with some physics experience. An interest in motor sport may be motivationally helpful – understanding race lines and rally/race driving techniques may add to the project.

Breakdown of Project with AI driven prompts and why I represent best person for Project: (AI MARKED IN RED)

1. Problem Context

Key influences:

- acceleration
- braking
- road geometry
- friction (skid)
- centre of gravity (rollover)

With freedom of racing line and points around the track defined as waypoints to get a snapshot of the car's telemetry. We aim to create a model that will find the limits of the track, follow initially traditional racing lines, and then find ways in which the car can optimise and try new things around a lap to optimise lap time as we move through new generations.

This data would be then given to a driver to give context clues as to areas in which more lap time may exist.

2. Core Idea

Represent the track as a spline (smooth curve) fitted to waypoints.

Track displayed as a spline to create a way in which the ai model can make its way through the circuit. I believe also this should be based on a basic real-life track to allow for context clues to be generated from video,

Model car physics to compute feasible speeds along this spline.

Physics should represent the key characteristics and telemetry in the car to assess the VMAX of the car through a corner; this can then be combined with other points and sector times to get a collective picture.

3. Key Steps in the Approach

Phase A – Setup

1. Platform choice

Options: Processing (Java-based visual tool), PyGame (Python + 2D graphics), OpenGL/WebGL (3D, more complex).

Simpler prototypes = PyGame/Processing.

I have experience within Processing, and I believe it could be used as a basic way of prototyping ideas and producing quick ways to improve the model in the initial stages of the project.

PyGame, I have no experience with, but I have coded lightly within Python, and I do have a good understanding of the concepts involved.

OpenGL, I have an affinity to work within, and I would love to justify a reason to jump in and generate 3D graphics.

2. Track definition

Waypoints from synthetic design (like an oval or “S” curve) or real data (e.g. OpenStreetMap).

Interpolation

Cubic splines or Bézier curves to get a continuous smooth track.

I believe that an interpretation of a basic Karting Circuit such as Buckmore Park in the UK or the Indy Circuit at Brands Hatch and it can be modelled quickly, and I have experience with the circuit for context cues.



Phase B – Physics Model

Maximum velocity per point

Use physics: centripetal force = road friction force

$$v_{max} = \mu \cdot g \cdot r$$

(where μ is friction coefficient, g gravity, r curvature radius).

Acceleration/braking

Equations of motion:

$$v^2 = u^2 + 2as$$

$$s = u \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

All these concepts and equations I have gotten experience within my time in TUD Mechanical Engineering Course, and I really enjoyed the modules. I have also used some of these formula's in CS171 within Processing.

Phase C – Path Optimization

- **Extend to full track width.**
Making sure that the Model utilizes the full limits of the track and keeps momentum.
- **Each point has possible horizontal offset (not just centre line).**
Points are horizontal across the track and are a sectorized zone which can be crossed at any point of the track.
- **Generate multiple paths.**
Give the model creativity to find alternate non traditional routes.
- **Sample different offsets to create candidate lines.**
Using Different offsets, the model can be encouraged the model to depart from traditional lines.
- **Genetic Algorithm (Fitness = lap time)**
The model will evolve based on the genetics' fitness i.e. Lap Time
- **Represent a path as a sequence of offsets (genes).**
A path that will show all the evolutions of the model over time
- **Use mutation/crossover to evolve better paths.**
Ability to change and combine model aspects to optimize models.

4. Skills Required

- **Programming: ability to code splines, physics equations, GA**
 - MH601 results are average of 67.93%.
(1st Year 67.6%, 2nd Year: 65.8%, 3rd Year 70.4%)
 - Great adaptability and strong ability to self-learn.
 - Experience in processing, java, C++, React
- **Maths/Physics: kinematics, circular motion, optimization.**
 - Studied Physics at Leaving Certificate Level
 - Pursued Level 7 in Mechanical Engineering
- **Motorsport knowledge: helps to anticipate realistic racing lines.**
 - Raced in Go Karting Leagues at National Level.
 - Racing enthusiast since early age with a good understanding of vehicle dynamics.
 - Worked at a Go karting Track for around 2 years and got familiar with different lines based on kart abilities.
 - Passion for topic matter

5. Complexity Analysis

Defining waypoints and splines.

Good Understanding of subject matter never applied it in practice; however, eager to learn.

Basic physics equations (straight-line motion).

Good Understanding of subject, experience in applying in Processing.

Accurately simulating braking/acceleration transitions.

Understanding of real-life philosophy of braking and acceleration transitions within motorsport. I know how it should be represented and would be excited to model it.

Implementing Genetic Algorithm efficiently.

I have a passion for machine learning alongside strategy within motorsport and its many applications especially within F1 and I would love to take this opportunity to work with the Genetic algorithm which is utilised widely in the industry.